



# Menopausal hot flashes

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Literature review current through: **Aug 2026**.

This topic last updated: **Jan 06, 2026**.

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## INTRODUCTION

Hot flashes occur in approximately 75 to 80 percent of menopausal women in the United States. The flashes most often begin in the perimenopausal period, although, in some women, they do not begin until after menopause. Hot flashes are almost always due to menopause. Other causes, including carcinoid syndrome, are uncommon ( [table 1](#)).

The prevalence, pathophysiology, and treatment of hot flashes will be reviewed here. Other menopausal symptoms and the risks, benefits, and practical aspects of menopausal hormone therapy (MHT) are reviewed separately. (See "[Menopause: Clinical features and diagnosis](#)" and "[Menopausal hormone therapy: Benefits and risks](#)" and "[Treatment of menopausal symptoms with hormone therapy](#)" and "[Genitourinary syndrome of menopause: Treatment of vulvovaginal atrophy](#)".)

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## EPIDEMIOLOGY

Symptoms become far more common during the menopausal transition, with a frequency of approximately 40 percent in the early transition and increasing to 60 to 80 percent in the late menopausal transition and early postmenopausal stage ( [figure 1](#)) [1-5].

**Prevalence and natural history** — Vasomotor symptoms (VMS) or "hot flashes" are the most common complaint during the menopausal transition, occurring in up to 80 percent of women [2,6-8], although the frequency appears to vary by culture and ethnicity [9,10] (see '[Risk factors](#)' below). Many to most women describe their symptoms as severe, but only approximately 20 to 30 percent of women seek medical attention for treatment [1,11,12].

Some women first develop hot flashes that cluster around menses during their late reproductive years, but symptoms are typically mild and do not require treatment.

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Four different trajectories of menopausal symptoms occur with nearly equal frequency: early onset 18.4 percent; onset near final menstrual period with later decline 29.0 percent; early onset with persistently high frequency 25.6 percent; and persistently low frequency 27.0 percent ( [figure 2](#)) [13].

### **Risk factors**

- Obesity – Obese postmenopausal women have higher serum [estrone](#) concentrations than lean women due to increased peripheral conversion of androstenedione in adipose tissue [14,15], but paradoxically, these women are more likely to have hot flashes [16,17]. Weight loss may help reduce their hot flashes [18]. (See '[Complementary and alternative therapies](#)' below.)
- Smoking [16,17].
- Reduced physical activity.
- Socioeconomic factors – Obtaining less than a high school education and having difficulty paying for basics are associated with a higher frequency of hot flashes [16].
- Hormonal concentrations – Annual serum follicle-stimulating hormone (FSH) levels, but not other hormones (estradiol, testosterone, dehydroepiandrosterone sulfate [DHEAS]), when collectively modeled longitudinally, are associated with both the prevalence and frequency of VMS [1].
- Ethnic factors – African American women report more frequent hot flashes than Caucasian women, and Japanese and Chinese women report less [8,16].

**Duration** — It had been thought that VMS diminishes and stops within a few years of onset in most women [12]. However, accumulating data suggest that symptoms typically last far longer [19-22]:

- In the Penn Ovarian Aging Study, the mean duration of moderate/severe hot flashes was 4.9 years, but one-third of women continued to experience moderate/severe hot flashes for 10 years after the final menstrual period (FMP) [21].
- Nine percent of women were still experiencing hot flashes at age 72 years in one study [20]. In a second study, 8 percent of women continued to have hot flashes into their late

menopausal years ( $\geq 20$  years beyond menopause) [19].

- The best estimate of total VMS duration comes from the Study of Women across the Nation (SWAN) [22]. Among 1449 women with VMS, the median total VMS duration was 7.4 years, with symptoms persisting for a median of 4.5 years after the FMP. Women who were premenopausal or early perimenopausal when they first experienced VMS had the longest total duration ( $>11.8$  years, post-FMP median duration 9.4 years). When compared with other racial/ethnic groups, African American women had the longest total VMS duration (10.1 years), while Chinese and Hispanic women had the shortest (approximately five years). The different trajectories of menopausal symptoms are described above ( [figure 2](#)). (See 'Prevalence and natural history' above.)

The long duration of hot flashes raises treatment challenges for many women (typically no more than four to five years to avoid excess breast cancer risk). Therefore, many women will continue to be symptomatic after their MHT is discontinued. (See "[Treatment of menopausal symptoms with hormone therapy](#)", section on 'Our preferred regimens'.)

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## PATHOPHYSIOLOGY

**Thermoregulatory dysfunction** — Hot flashes are mediated by thermoregulatory dysfunction at the level of the hypothalamus and are induced by estrogen withdrawal [23]. The thermoneutral zone is narrowed in women with hot flashes [24]. While premenopausal women initiate mechanisms to dissipate heat when the core body temperature increases by  $0.4^{\circ}\text{C}$ , this happens with much lower increases in temperature in menopausal women [25].

The feeling of warmth results from inappropriate peripheral vasodilatation with increased digital and cutaneous blood flow. Perspiration results in rapid heat loss and a decrease in core body temperature below normal. Shivering may then occur as a normal mechanism to restore the core temperature to normal [23]. Estrogen administration restores the "thermoneutral zone" to normal and largely abolishes hot flashes.

**Role of hypoestrogenemia** — The observation that prepubertal girls, whose endogenous estrogen levels are very low, do not experience hot flashes provides evidence that hypoestrogenemia alone is not adequate for hot flashes to occur. Data from patients with Turner syndrome and functional hypothalamic amenorrhea suggest that exposure to adult levels of estrogen followed by withdrawal are necessary for hot flashes to occur. (See "[Functional hypothalamic amenorrhea: Pathophysiology and clinical manifestations](#)", section on 'Clinical manifestations'.)

Evidence for central mediation of the changes in temperature comes from studies demonstrating that hot flashes occur simultaneously with, but are not caused by, pulses of luteinizing hormone (LH) [26,27]. The close proximity of the medial preoptic area (the major

thermoregulatory area in mammals) and a high density of gonadotropin-releasing hormone (GnRH)-containing neurons suggest the possibility of simultaneous activation of a burst of GnRH release and thermoregulatory changes.

**Neuroendocrine and genetic basis** — One hypothesis, now supported by increasing evidence, suggests that KNDy (kisspeptin, neurokinin B, and dynorphin) neurons mediate the LH and thermoregulatory changes [28] observed in menopausal women. Estrogen withdrawal affects hypothalamic **KNDy neurons**, causing dysregulated thermoregulation that results in hot flashes and night sweats. Neurokinin B (NKB) (a hypothalamic neuropeptide) and its receptor (NK3R) are also thought to be associated with vasomotor symptoms as *TACR3* is the gene that encodes NK3R, and peripheral infusion of NKB induces hot flashes in postmenopausal women [29]. In addition, women who have variations in the gene that codes for tachykinin receptor 3 (*TACR3*) are more likely to experience hot flashes than women without those variations [30]. (See '[Neurokinin 3 receptor antagonists](#)' below.)

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## TYPICAL FEATURES

Hot flashes typically begin as the sudden sensation of heat centered on the upper chest and face that rapidly becomes generalized. The sensation of heat often lasts from two to four minutes, is often associated with profuse perspiration and occasionally palpitations, and is sometimes followed by chills, shivering, and a feeling of anxiety. Hot flashes may range from an average of less than one each day to as many as one per hour during the day and night.

Hot flashes have a negative impact on quality of life as well as an association with adverse health indicators (increased cardiovascular risk and greater bone loss/increased bone turnover) [31,32].

Although several grading scales have been reported, a practical classification of hot flashes and night sweats utilizes a four-point scale [33]:

- Not present.
- Mild – Present but not interfering with sleep, daily activities, or work.
- Moderate – Interfere somewhat with sleep and daily activities.
- Severe – More frequent, severe, and bothersome hot flashes that interfere with sleep, daily activities, and work. So bothersome that usual activities cannot be performed.

It had been thought that vasomotor symptoms diminish and stop within a few years of onset in most women. However, vasomotor symptoms can persist for as long as 20 years past the final menstrual period (FMP) and may be variable between racial/ethnic groups [34]. (See '[Menopause: Clinical features and diagnosis](#)', section on 'Hot flashes'.)

**Other causes of sweating** — Other causes of sweating may mimic hot flashes or night sweats, including the carcinoid syndrome, medications, hyperthyroidism, infection, and malignancy, all of which are reviewed in detail separately ( [table 2](#) and [table 3](#)). (See "[Evaluation of the patient with night sweats or generalized hyperhidrosis](#)".)

**Effect on sleep** — When hot flashes occur at night, women typically describe them as "night sweats." Vasomotor symptoms (VMS) at night can cause significant sleep disturbances in many women [35-37]. In one study of 34 symptomatic perimenopausal women undergoing one to five overnight polysomnographic (PSG) recordings, 69 percent of hot flashes were associated with awakening [38]. Hot flashes, although not the only cause of sleep disturbance, accounted for 27 percent of wakefulness during the night in these women. The estimated prevalence of difficulty sleeping based upon two longitudinal cohort studies was 32 to 40 percent in the early menopausal transition, increasing to 38 to 46 percent in the late transition [7,39].

PSG studies suggest that nocturnal hot flashes are more common during the first four hours of sleep, whereas rapid eye movement (REM) sleep in the subsequent four hours suppresses hot flashes, arousals, and awakenings [35].

**Other sleep problems** — VMS can disrupt sleep in peri- and postmenopausal women, but there are many other issues that disturb sleep in this population including primary sleep disorders, anxiety, and depression. Menopausal sleep disturbances are discussed separately. (See "[Menopause: Clinical features and diagnosis](#)", section on 'Sleep disturbance'.)

Women with seven or more daily moderate to severe VMS (defined as with sweating or affecting function) report interference with sleep (94 percent), concentration (84 percent), mood (69 percent) and sexual activity (40 percent) [40]. The more severe the VMS, the more daily activities are impacted.

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## MENOPAUSAL HORMONE THERAPY

**General principles** — The approach to management of hot flashes is based upon a number of factors, including:

- The patient's assessment of her hot flashes and their impact on function.
- Medical history – Is the patient a candidate for menopausal hormone therapy (MHT)? (See "[Treatment of menopausal symptoms with hormone therapy](#)", section on 'Is my patient a candidate for MHT?'.)
- Personal choice – Is the patient interested in MHT?

- Coexistence of other menopausal symptoms, such as depression, as these women often require treatment with both MHT and antidepressants (usually selective serotonin reuptake inhibitors [SSRIs]). (See ["Treatment of menopausal symptoms with hormone therapy", section on 'Perimenopausal mood disorders'](#).)

**Women with mild symptoms** — Women with mild flashes (hot flashes that do not interfere with sleep, daily activities, or work) may not seek pharmacotherapy (see ["Typical features"](#) above). However, many do eventually seek help if symptoms worsen.

In the past, behavioral measures such as lowering room temperature, using fans, dressing in layers of clothing that can be easily shed, and avoiding triggers (such as spicy foods and stressful situations) were recommended. However, the efficacy of these approaches is not well established, and the Menopause Society does not recommend them [41].

Some clinicians recommend [vitamin E](#) to women with mild hot flashes because, at low doses, it is well-tolerated and not associated with toxicity [42]. In one report, vitamin E and placebo were associated with reductions in hot flash frequencies of 32 and 29 percent, respectively, a marginal difference that is not clinically important [43].

Other nonhormonal options that are often used but not recommended by the Menopause Society include paced respiration, supplements/herbal remedies, exercise, yoga, mindfulness-based intervention, relaxation, soy, cannabinoids, acupuncture, and [clonidine](#) [41].

**Moderate to severe hot flashes** — Hormonal or other pharmacotherapy is usually needed for women with more bothersome hot flashes. For most women with moderate to very severe hot flashes and no contraindications, we suggest menopausal hormone therapy (MHT). MHT is the broad term used to describe both unopposed estrogen use for women who have undergone hysterectomy and estrogen combined with progestogen therapy to prevent estrogen-associated endometrial hyperplasia in women with an intact uterus.

The management of menopausal women who choose to take hormone therapy and the available estrogen and progestin preparations are discussed in detail separately. (See ["Treatment of menopausal symptoms with hormone therapy"](#) and ["Preparations for menopausal hormone therapy"](#) and ["Menopausal hormone therapy: Benefits and risks"](#).)

The majority of perimenopausal and recently menopausal women are good candidates for hormone therapy for symptom relief. However, for women with a history of breast cancer, coronary heart disease (CHD), a previous venous thromboembolic event (VTE) or stroke, or those at moderate or high risk for these complications, alternatives to hormone therapy should be suggested [44,45]. (See ["Special considerations"](#) below and ["Treatment of menopausal symptoms with hormone therapy", section on 'Is my patient a candidate for MHT?'](#).)

A more detailed discussion of MHT, including guidance on the optimal type, dose, route, and duration of therapy, approaches for those who are unable to tolerate oral progestogens, and treatment options we do not recommend (compounded bioidentical hormone therapy), is found separately. (See ["Treatment of menopausal symptoms with hormone therapy", section on 'Our preferred regimens'](#).)

**Tibolone** — [Tibolone](#) is a synthetic steroid whose metabolites have estrogenic, androgenic, and progestogenic properties. Widely used in Europe and other countries for nearly 20 years, it reduces hot flashes, has a beneficial effect on bone mineral density [46-49], and may be more effective than estrogen/progestin therapy for treatment of sexual dysfunction in postmenopausal women. However, tibolone has been associated with an increased risk of stroke and possibly breast cancer and myocardial infarction. We therefore do not recommend its routine use for hot flash management. Additional information on tibolone is found elsewhere. (See ["Preparations for menopausal hormone therapy"](#) and ["Overview of sexual dysfunction in females: Management"](#), section on 'Tibolone' and ["Overview of the management of low bone mass and osteoporosis in postmenopausal women"](#), section on 'Therapies not recommended'.)

**Not recommended: Compounded bioidentical hormone therapy** — The term "bioidentical hormone" technically refers to an exogenous hormone with the same molecular structure as a hormone that is endogenously produced. However, in popular culture, the term refers to the use of custom-compounded, multihormone regimens with dose adjustments based upon serial hormone monitoring. Some postmenopausal women have turned to this approach based upon claims of better safety and efficacy. However, there is no evidence to support this, and we suggest against this approach [34]. (See ["Preparations for menopausal hormone therapy"](#), section on 'Compounded bioidentical hormone therapy'.)

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## NONHORMONAL PHARMACOTHERAPY

Hormone therapy remains the most effective treatment for menopausal symptoms. However, for women with moderate to severe hot flashes who are not candidates for hormone therapy or who choose not to take it, we suggest nonhormonal agents [41,44].

The best studied agents with positive results include selective serotonin reuptake inhibitors (SSRIs), serotonin-norepinephrine reuptake inhibitors (SNRIs), antiepileptics, [gabapentin](#), [clonidine](#), [oxybutynin](#), and other centrally acting drugs.

These drugs are also often tried in women who experience recurrent hot flashes after stopping menopausal hormone therapy (MHT). Our choice of drug depends upon whether the patient is taking [tamoxifen](#), the pattern of hot flashes, and the presence of a mood disorder or sleep problem.

Two neurokinin 3 receptor antagonists have been approved and are available for clinical use. (See '[Neurokinin 3 receptor antagonists](#)' below.)

## Menopause Society recommendations

As noted, menopausal hormone therapy (MHT) is the most effective option for management of menopausal symptoms. But some women have contraindications to its use and others choose not to take it. The 2023 nonhormonal therapy position statement from the Menopause Society suggests a number of treatment options for hot flashes in women who cannot take hormone therapy [41]. These include cognitive-behavioral therapy, SSRIs, SNRIs, [oxybutynin](#), [gabapentin](#), [fezolinetant](#), and [elinzanetant](#).

The Society also identified nonhormonal therapies that they do not recommend [41]. (See '[Not recommended](#)' below.)

**Placebo effect** — High placebo effects are seen in a number of conditions such as migraines, irritable bowel syndrome, asthma, and others [50].

There is also an important placebo effect (as high as 20 to 50 percent) in clinical trials of nonhormonal therapies for hot flashes [51-53]. The high placebo rates highlight the importance of testing every hot flash therapy against placebo in randomized, controlled trials.

**SSRIs/SNRIs** — The most effective pharmacologic alternatives to estrogen include some of the antidepressants in the selective serotonin reuptake inhibitor (SSRI) and serotonin-norepinephrine reuptake inhibitor (SNRI) classes [51,52,54,55]. Their efficacy has been demonstrated in individual placebo-controlled trials [56-65], a pooled analysis of seven trials of SSRIs/SNRIs (and three trials of [gabapentin](#)) [51], and a meta-analysis of 43 trials of nonestrogen therapies [52]. Key points about the use of these drugs for VMS in postmenopausal women include the following:

- **Rapid response** – The clinical response is more rapid (days) than the typical response to SSRIs for depression (weeks) [63].
- **Patient population** – These drugs appear to be equally effective in women with breast cancer (with or without [tamoxifen](#)) and in women with surgical or natural menopause [66].
- **Efficacy** – Although no head-to-head trials have been performed, indirect comparisons suggest that SSRIs ([citalopram](#), [escitalopram](#), [paroxetine](#)) and SNRIs ([venlafaxine](#), and [desvenlafaxine](#)) have a similar modest benefit for hot flashes. Results from some of the individual trials of nonestrogenic treatment options are shown in the figure ( [figure 3](#)) [57,60,61,67].
- **Choice of drug** – We typically start with an SSRI ([citalopram](#) or [escitalopram](#)) rather than an SNRI to minimize side effects and withdrawal symptoms [62,63,68-70].

As [paroxetine](#) blocks the conversion of [tamoxifen](#) to active metabolites through *CYP2D6*, it should not be used in women on tamoxifen. The impact of *CYP2D6* inhibition on tamoxifen efficacy is a subject of substantial controversy with no consensus at present [71]. (See '[Women with breast cancer](#)' below.)

We do not prescribe [sertraline](#) or [fluoxetine](#), as neither has a clinically important effect on hot flashes. In three trials, sertraline was no more effective than placebo [58,68,72,73].

While [venlafaxine](#) is well studied, it has more acute toxicity (nausea and vomiting) and can have significant withdrawal symptoms in occasional patients [56,59,74,75]. If it is used, we suggest the sustained-release preparation starting with 37.5 mg/day for one week, increasing to 75 mg/day after the first week, to reduce the incidence of initial nausea [59]. When patients stop therapy, they should be instructed to decrease to 37.5 mg for one week before stopping altogether.

One trial reported that [venlafaxine](#) (75 mg/day) had similar efficacy for hot flash relief as a low dose of oral estradiol (0.5 mg) [74]. However, in a clinical setting, increasing the dose of venlafaxine would be unlikely to result in a further reduction in hot flashes, while increasing the dose of estrogen could potentially eliminate hot flashes altogether.

[Desvenlafaxine](#), the main active metabolite of [venlafaxine](#), has similar benefits for hot flashes and a similar side effect profile as venlafaxine [76]. While not commonly used, a potential advantage of this agent is that dose escalation is not usually necessary, as the variability in conversion of precursor to active drug is not a factor.

- Since hot flashes gradually subside without therapy in most, but not all, postmenopausal women, any drug can be eventually tapered. Women on SSRIs or SNRIs should taper the drug gradually to avoid drug withdrawal symptoms. Hot flashes often persist for 10 years or more, so many women need to resume treatment or try another option for symptoms management.
- For women taking nonhormonal pharmacotherapies, we suggest trying another pharmacologic agent if one therapy is ineffective or not tolerated. If one SSRI/SNRI is tried but does not work, a second one could be tried before moving to another class of drug. If the SSRI/SNRI class of drugs is ineffective, a trial of [gabapentin](#) would be reasonable. Other options would include [pregabalin](#) or [clonidine](#). (See '[Gabapentin](#)' below.)
- If these pharmacologic options or menopausal hormone therapy (MHT) are ineffective or not tolerated, other options can be explored (see '[Complementary and alternative therapies](#)' below). In addition, other causes of flushes or night sweats should be considered in women who do not respond to multiple therapies ( [table 2](#) and [table 3](#)).

## Gabapentin

- **Gabapentin** – [Gabapentin](#) is an antiseizure medication that is also used for other indications, including post-herpetic neuralgia and painful diabetic neuropathy. However, it is also effective for hot flashes in some women. (See "[Antiseizure medications: Mechanism of action, pharmacology, and adverse effects](#)" and "[Overview of the management of epilepsy in adults](#)".)

[Gabapentin](#) 900 mg/day (typically administered as 300 mg three times per day) has been shown to be more effective than placebo in a number of trials ( [figure 3](#)) [51,64,65,77,78].

Nocturnal hot flashes appear to occur primarily in the first four hours of sleep, whereas rapid eye movement (REM) sleep in the subsequent four hours suppresses hot flashes [35]. In women whose hot flashes occur primarily at night, the authors have found that a single dose of [gabapentin](#) given at bedtime often relieves the hot flashes that awaken patients from sleep. If night sweats do occur, many women find that they fall back asleep more easily than they did pretreatment. The sedating effect of gabapentin and its effects to reduce vasomotor instability work together. This regimen also minimizes the side effects of gabapentin encountered during the daytime.

Individualized dosing is needed because of variable patient sensitivity to this agent. We suggest starting with 100 mg one hour before bedtime and increasing by 100 mg every three nights until relief of hot flashes, side effects, or a maximum of 900 mg.

At much higher doses, [gabapentin](#) may be as effective as estrogen, but it is limited by its side effects. In a one trial (20 patients per arm) of women receiving gabapentin (dose titrated up to 2400 mg over 12 days), [conjugated estrogen](#) (0.625 mg), or placebo, both gabapentin and estrogen reduced hot flash severity score to a similar degree when compared with placebo (72, 71, and 54 percent, respectively) [79]. However, at this dose, gabapentin caused significant side effects including headache, dizziness, and disorientation.

Combined therapy with [gabapentin](#) and an antidepressant (SSRI or [venlafaxine](#)) does not appear to be more effective for hot flashes than gabapentin alone in patients with inadequate control of their hot flashes with an antidepressant alone [80]. Nonetheless, when switching from an antidepressant to gabapentin, it might be best to continue the antidepressant for the first two weeks and then titrate down. The reason for this is that there were more side effects noted in the patients who stopped the antidepressant than in those who took both. This is likely from the withdrawal symptoms from discontinuing the antidepressant, which the patient might inappropriately attribute to the newly initiated gabapentin.

- **Pregabalin** – [Pregabalin](#), like [gabapentin](#), is used for a number of indications, including seizures, neuropathic pain, and postherpetic neuralgia, and is effective in relieving hot flashes. The drug at the recommended dose of 300 mg is reasonably well tolerated. We

currently suggest gabapentin over pregabalin as it has been better studied for hot flashes and is less expensive [81].

- **Oxybutynin** – [Oxybutynin](#), an oral anticholinergic agent used primarily for overactive bladder, was more effective than placebo for relief of hot flashes in an observational study, a randomized clinical trial, and a preliminary report from a second clinical trial [82]. The optimal daily dose appears to be 5 to 10 mg [82]. Dry mouth, the most common side effect of oxybutynin, occurred in 20 to 30 percent of subjects in one trial [82] and 50 percent in another [83]. Further studies are required to determine its relative efficacy compared with other agents.

Observational studies suggest that [oxybutynin](#) and other anticholinergic drugs may be associated with an increased risk of dementia in older adults. (See "[Risk factors for cognitive decline and dementia](#)", section on 'Medications'.)

Although [oxybutynin](#) is suggested by the Menopause Society, we use it infrequently because of its side effects and possible risk of dementia.

**Neurokinin 3 receptor antagonists** — Neurokinin 3 receptor (NK3R) antagonists represent another nonhormonal approach to the management of hot flashes. The development of NK3R antagonists is based upon the observation that the thermoregulatory center in the hypothalamus is innervated by kisspeptin/neurokinin B/dynorphin (KNDy) neurons that are stimulated by neurokinin B (NKB) and inhibited by estrogen. After menopause, estrogen declines and NKB signaling is increased, resulting in unregulated KNDy neuron activation and vasomotor symptoms [29,30,84].

The first NK3R antagonist to be approved and available for clinical use is [fezolinetant](#). In one trial of over 500 postmenopausal women with moderate-to-severe hot flashes, fezolinetant 30 mg/day or 45 mg/day significantly reduced hot flash frequency and severity when compared with placebo [85]. After 12 weeks of therapy, the mean reductions in hot flash frequency for fezolinetant 45 mg, 30 mg, or placebo were 64, 59, and 45 percent, respectively. Women receiving fezolinetant 45 mg also had significant improvements in sleep disturbances when compared with placebo. Similar results were seen in the second fezolinetant phase 3 trial [86].

Adverse event rates, including liver enzyme abnormalities, were <2 percent in all groups [85]. However, based upon liver-related adverse events in earlier trials as well as cases of serious drug-induced hepatotoxicity in the postmarketing setting, the FDA now requires measurement of serum transaminases (AST, ALT) and bilirubin at baseline, monthly for the first three months, and then at six and nine months after initiation [87]. [Fezolinetant](#) should not be started if any result is  $\geq 2$  times the upper limit of normal (ULN).

In addition, the drug should be discontinued for any symptoms or signs suggesting liver injury (nausea, vomiting, jaundice, pruritus, etc) or if transaminases are >5 times ULN or if

transaminases are >3 times ULN and total bilirubin is >2 times ULN.

[Elinzanetant](#), a dual NK3/NK1 receptor antagonist also effectively reduces both hot flash frequency and severity [88]. The daily dose is 120 mg (two 60 mg capsules daily) and baseline liver function tests are recommended as they are for [fezolinetant](#) [89].

In addition, [elinzanetant](#) is effective for hot flashes due to endocrine therapy for breast cancer [90].

**Not recommended** — Nonhormonal therapies not recommended by the Menopause Society include paced respiration, supplements/herbal remedies, cooling techniques, avoiding triggers, exercise, yoga, mindfulness-based intervention, relaxation, soy, cannabinoids, acupuncture and [clonidine](#).

[Clonidine](#), a centrally active alpha-2 adrenergic agonist, was modestly more effective than placebo in a meta-analysis of 10 trials [52]. However, clonidine is now infrequently used and not recommended by the Menopause Society given its side-effect profile (dry mouth, dizziness, constipation, and sedation) and the availability of other effective nonhormonal medications for hot flashes. If used (ie, if all other options are ineffective or not tolerated), the transdermal preparations might be superior to oral tablets because of stable blood levels. The suggested starting dose for the TTS-1 (transdermal therapeutic system 1) patch, is 0.1 mg per day, and increasing to TTS-2 (0.2 mg/day) and TTS-3 (0.3 mg/day) based on efficacy and side effects. Oral clonidine, at a dose of 0.1 to 1 mg/day in divided doses, may also be used (and may be substantially less expensive). However, clonidine is not recommended by the Menopause Society [41].

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## SPECIAL CONSIDERATIONS

**Women with breast cancer** — Women with breast cancer have more problems with hot flashes than other women for a number of reasons. One is that premenopausal women may become estrogen deficient relatively abruptly, due to chemotherapy that results in early ovarian failure or due to bilateral oophorectomy. In addition, many postmenopausal women with breast cancer receive [tamoxifen](#), an adjuvant therapy associated with hot flashes in up to 80 percent of women, 30 percent of whom rate them as severe. Women who take tamoxifen for chemoprevention may also experience severe hot flashes. Inheritance of specific estrogen receptor genotypes may also influence the risk and severity of tamoxifen-induced hot flashes [91]. (See "[Managing the side effects of tamoxifen and aromatase inhibitors](#)", section on 'Hot flashes'.)

Aromatase inhibitors may also cause hot flashes, but they tend to be less frequent and less severe [92]. Lastly, in women with a history of breast cancer, systemic estrogen therapies are considered relatively contraindicated.

The principles and approach to treating women with breast cancer and hot flashes include [93]:

- The efficacy of the nonhormonal therapies described above for hot flashes (including SSRIs/SNRIs) appears to be similar in women taking [tamoxifen](#) compared with those who are not taking tamoxifen [66].
- While SSRIs help reduce hot flashes in women on [tamoxifen](#), coadministration of some of these drugs also inhibits the activity of *CYP2D6*, the enzyme that converts tamoxifen to its most active metabolite, endoxifen. Among SSRIs and SNRIs used to treat hot flashes, there is a gradient of potency for inhibition of *CYP2D6*; for example, [paroxetine](#) is a strong *CYP2D6* inhibitor, while [venlafaxine](#), [citalopram](#), and [duloxetine](#) are less so [94]. While there is no evidence for an increased risk of breast cancer recurrence in women taking strong inhibitors of *CYP2D6* (paroxetine) with tamoxifen, we suggest other agents such as citalopram or venlafaxine in this setting.
- SSRIs do not interfere with the metabolism of aromatase inhibitors and can therefore be used safely in women receiving adjuvant aromatase inhibitor therapy.
- We suggest using an SSRI or an SNRI for women on [tamoxifen](#) who are experiencing moderate to severe hot flashes throughout the day and night (with the exception of the potent *CYP2D6* inhibitors [paroxetine](#) and [fluoxetine](#)). The largest experience is with [venlafaxine](#), but [citalopram](#), [desvenlafaxine](#), and [escitalopram](#) are also reasonable choices. We typically use citalopram or escitalopram as they are better tolerated than venlafaxine.
- Neurokinin B receptor antagonists are a novel class of drugs for hot flashes. [Fezolinetant](#) and [elinzanetant](#) are now available. (See '[Neurokinin 3 receptor antagonists](#)' above.)
- In women whose hot flashes are particularly symptomatic at night, graded doses of [gabapentin](#) can be administered at bedtime. (See '[Gabapentin](#)' above.)

**Perimenopausal women** — Women often seek intervention for hot flashes while they are still having menstrual cycles, ie, during the late menopausal transition. Management of these patients with MHT can be more challenging; a low-dose oral contraceptive is often used to control bleeding and provide contraception. This issue is reviewed in more detail separately. (See "[Treatment of menopausal symptoms with hormone therapy](#)", section on '[Use of oral contraceptives during the menopausal transition](#)'.)

**Coexisting depression** — Women with significant hot flashes and coexisting depression often require treatment with both MHT and antidepressants (usually selective SSRIs). (See "[Treatment of menopausal symptoms with hormone therapy](#)", section on '[Perimenopausal mood disorders](#)'.)

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## COMPLEMENTARY AND ALTERNATIVE THERAPIES

It is estimated that between 50 to 75 percent of postmenopausal women use complementary and alternative therapies for management of menopausal symptoms [95,96], and the prevalence may be even higher in breast cancer patients [95]. For most of these therapies, safety and efficacy are not well established, but selected individual studies report benefit. Interpretation of the available data has been difficult given the small size and short duration of many studies. As placebos reduce hot flashes in up to 50 percent of patients, carefully controlled, randomized trials are necessary to demonstrate the efficacy of specific agents.

As noted, complementary therapies that are not recommended by the Menopause Society include paced respiration, supplements/herbal remedies, cooling techniques, avoiding triggers, exercise, yoga, mindfulness-based intervention, relaxation, soy, cannabinoids, and acupuncture [41,95].

- **Cognitive behavioral therapy (CBT)** – CBT appears to be a modestly effective intervention for menopause-associated insomnia [97] but less so for hot flashes. A major limitation of CBT is the need for frequent in-person visits.

Two randomized trials of intensive group or self-help CBT in breast cancer survivors [98] and peri- and postmenopausal women without breast cancer [99] demonstrated a significant reduction of bother related to hot flashes compared with those receiving no treatment. However, there was no reduction in hot flash frequency.

New-onset sleep disturbances are common during perimenopause and early menopause, typically in women with hot flashes, but also in women without hot flashes (see 'Effect on sleep' above). Women with combined VMS and insomnia have lower physical and mental quality of life than women without insomnia, and their insomnia symptoms improve with MHT [100,101]. Many women are treated with prescription sleeping pills, but a nonpharmacologic alternative (in-person CBT for insomnia [CBT-I]) is also effective (albeit with the practical limitation of frequent in-person visits).

In a trial of **telephone-based** therapies, 106 peri- and postmenopausal women with insomnia and hot flashes were randomly assigned to an eight-week intervention with CBT-I versus standard menopause education control (MEC) [97]. Insomnia severity was assessed with the Insomnia Severity Index (ISI), a seven-item questionnaire. After eight weeks, CBT-I, but not MEC, resulted in a clinically meaningful insomnia score reduction. The percentage of women who achieved ISI scores in the "no insomnia" range was 70 and 24 percent for CBT-I and MEC, respectively. CBT-I reduced hot flash "bother" but not frequency.

- **Hypnosis** – Hypnosis may be beneficial for hot flashes, but data are limited [102,103]. In a trial comparing hypnosis to a "structured attention intervention," hypnosis was more

effective. At 12 weeks of follow-up, the mean reduction in physiologically monitored hot flashes per day was six (57 percent) in the hypnosis group versus one (10 percent) in the control group. While these results appear promising, the response rate in the control group is unusually low; a placebo effect of 20 to 50 percent is typically seen in hot flash trials. Before hypnosis can be recommended, procedures need to be standardized and educational tools developed to facilitate development of competency with this method. (See ['Nonhormonal pharmacotherapy'](#) above.)

- **Mind-body-based therapies** – Small numbers of studies of mind-body-based therapies (stress management, relaxation, deep breathing techniques, and guided imagery) had mixed results in two meta-analyses [[54,104](#)]. A 2014 systematic review of relaxation studies reported no differences between relaxation techniques, paced respiration, placebo, and no treatment [[105](#)].
- **Mindfulness training** – A randomized trial of mindfulness-based stress reduction (MBSR) reported a decrease in bother from menopausal hot flashes but no effect on hot flash intensity scores (based upon frequency and severity) when compared with no treatment [[106](#)]. As MBSR requires intensive training like CBT, further evidence of benefit is needed before routinely recommending this approach.
- **Stellate ganglion block** – Pilot trials suggested that a stellate ganglion block can alleviate hot flashes. In a trial of 40 women with moderate to severe hot flashes undergoing stellate ganglion block or sham injection, the overall frequency of hot flashes in the subsequent six months was not different between groups, but the frequency of moderate to very severe symptoms was reduced more in the active group compared with the sham treatment group (event rate ratio 0.50, 95% CI 0.35-0.71) [[107](#)]. In a second trial of 40 patients randomized to three months of stellate ganglion blocks or [pregabalin](#) therapy, both therapies were effective, but the reduction in hot flashes was greater with stellate ganglion blocks [[108](#)].
- **Plant-based therapies** – There are a variety of plant-based therapies used for hot flashes including isoflavones (dietary or supplements) and herbal therapies.

Isoflavones and phytoestrogens (nonsteroidal compounds that occur naturally in many plants, fruits, and vegetables) have not been consistently found to be more efficacious than placebo for hot flashes. Phytoestrogens have both estrogenic and antiestrogenic properties and are categorized as isoflavones, coumestans, or lignans.

Two types of isoflavones, genistein and daidzein, are found in soybeans, chickpeas, and lentils and are thought to be the most potent estrogens of the phytoestrogens (although they are much weaker than human estrogens). Lignans (eg, enterolactone and enterodiols) are found in flaxseed, lentils, grains, fruits, and vegetables. Most trials have not shown benefit for hot flashes when compared with placebo [[109-115](#)]. In a review of 11

randomized clinical trials of soy or isoflavone supplementation, only three of eight trials with at least six weeks of follow-up demonstrated a beneficial effect [109].

In contrast, a meta-analysis of 21 trials in menopausal women with hot flashes receiving phytoestrogens (dietary soy isoflavones, supplements of soy isoflavones, or red clover) reported a decrease in the number of daily hot flashes (-1.31, 95% CI -2.02 to -0.61) but no change in night sweat frequency [116]. Although this was statistically significant, it may not be clinically significant for women with severe, frequent hot flashes [117]. When different isoflavones were analyzed separately, red clover was ineffective for hot flashes, similar to results from previous studies [52,104,118].

A 2013 systematic review of 43 clinical trials found no beneficial effect of phytoestrogens of any type on hot flashes, with the exception of genistein [119]. Results of four genistein trials (not pooled for meta-analysis, because of high risk of bias) suggested that doses >30 mg/day might reduce hot flash frequency compared with placebo.

An issue of potential concern is that phytoestrogens are SERMs and therefore have both estrogen agonist and antagonist effects. However, the impact of increasing dietary soy or taking concentrated isoflavone supplements on breast cancer risk or recurrence in women with estrogen receptor-positive (ER+) breast cancer is unknown. Many experts suggest that dietary soy is reasonable in women with ER+ breast cancer but that dietary supplements should be avoided until their safety has been established.

- **Herbal therapies**

- **Black cohosh** – Among the alternative therapies available for management of hot flashes, black cohosh (*Actaea racemosa* or *Cimicifuga racemosa*) is one of the most widely used [95,96]. Although individual small, short-term trials suggested benefits from black cohosh [120,121], most have not [104,122-124]. In addition, systematic reviews and meta-analyses suggest that black cohosh is no more effective than placebo for treatment of hot flashes [116,125].

A potential safety concern about black cohosh is its possible estrogenic effect on the breast, but there is no evidence to date that use of black cohosh is associated with increased risk of recurrence in women with ER+ breast cancer [126,127]. However, some groups suggest that women with breast cancer wait until more safety data are available before using it [128].

There have also been concerns about possible hepatotoxicity (abnormal liver function tests) with black cohosh [129], but a meta-analysis of five black cohosh trials involving a total of 1117 women found no evidence of an adverse effect on liver function [130].

- **Chinese herbs** – Data from two meta-analyses suggest that Chinese herbal therapy is ineffective for the treatment of menopausal hot flashes [116,131].
- **Paced respiration** – A relaxation-based method known as paced respiration appears to be relatively easy to learn, but its impact on hot flash symptoms is not yet established [132,133].
- **Weight loss** – As noted above, obesity is a risk factor for hot flashes (see 'Risk factors' above). Weight loss may help reduce hot flashes, as illustrated by a six-month trial in overweight and obese pre- and postmenopausal women randomized to an intensive behavioral weight loss intervention or a structured health education program [18]. Among women with bothersome hot flashes at baseline, the intensive intervention resulted in a qualitative improvement in hot flashes when compared with the control intervention.
- **Acupuncture** – Acupuncture is among the most frequently used complementary therapies for hot flashes, but results thus far are conflicting [134-138]. In a systematic review and meta-analysis, acupuncture was less effective than MHT (three trials), no different from sham acupuncture (eight studies), but statistically more effective than no therapy (four trials) [139].

Results from individual sham acupuncture (eg, placebo-controlled) trials have been variable; most are limited by small sample size and methodologic flaws. The best evidence to date that acupuncture is no more effective than placebo (sham acupuncture) comes from a randomized, sham-controlled trial in 327 peri- or postmenopausal women with moderate to severe hot flashes who were randomly assigned to 10 traditional Chinese acupuncture or noninsertive sham acupuncture treatments over eight weeks [140]. Hot flash scores, a calculated score based upon hot flash frequency and severity, were no different between the groups at the end of treatment (both showed approximately 40 percent improvement). Thus, like other nonhormonal and complementary therapies for hot flashes, acupuncture has an important placebo effect, but it has no additional benefit over sham acupuncture. (See 'Placebo effect' above.)

Similar results were seen in another trial [141]. Although electroacupuncture appeared to be as effective as [gabapentin](#) for reducing hot flashes, it was no better than sham acupuncture. (See 'Placebo effect' above.)

- **Evening primrose oil** – Evening primrose oil is a rich source of gamma-linolenic acid and is widely used by postmenopausal women to treat hot flashes. However, available data suggest that it is no more effective than placebo. In the only randomized trial to address this issue, 56 symptomatic postmenopausal women were randomly assigned to evening primrose oil (4 g/day plus 80 mg [vitamin E](#)) versus placebo for six months [142]. Among the 35 women who completed the trial, all showed improvement in their symptoms but no significant differences were found between the two groups.

- **Flaxseed** – Flaxseed has not been shown to be more effective than placebo for hot flashes [143,144].
- **Exercise** – Clinical trials, systematic reviews, and a meta-analysis have not found a significant beneficial effect of exercise on hot flashes [104,145,146]. Hormone therapy was more effective than exercise in one trial [147]. One possible explanation for lack of benefit is that exercise raises core body temperature, thereby triggering hot flashes.
- **Other** – A number of other herbal and complementary therapies have been studied with largely negative results. In a systematic review of 10 clinical trials noted above [109], no reduction in frequency of flushes was seen with ginseng or dong quai. Wild yam and progesterone creams also appear to be ineffective [104,109]. Traditional medicinal Chinese herbs, reflexology, and magnetic devices have all been studied and appear to have no beneficial effect [104].

For women who choose estrogen or combined estrogen-progestin therapies, short-term use has been suggested (generally for not more than five years) based on the original interpretation of the Women's Health Initiative (WHI) data (see "[Menopausal hormone therapy: Benefits and risks](#)"). However, several expert societies, including the Endocrine Society, the American College of Obstetrics and Gynecology, the Menopause Society, and the International Menopause Society no longer recommend a specific duration of treatment in symptomatic women, based upon revised safety data and the observation that hot flashes persist well beyond five years in many women ( [figure 2](#)). (See '[Duration](#)' above and "[Treatment of menopausal symptoms with hormone therapy](#)", section on '[Extended use of MHT](#)' and "[Menopausal hormone therapy: Benefits and risks](#)", section on '[Women 50 to 59 years](#)'.)

For select women who experience recurrent, bothersome hot flashes (unresponsive to nonhormonal therapies) after stopping estrogen, extended use of estrogen therapy can be considered for quality of life. (See "[Treatment of menopausal symptoms with hormone therapy](#)", section on '[Extended use of MHT](#)'.)

Some experts now suggest that menopausal hormone therapy can be considered for primary prevention of disease (cardiovascular disease, osteoporosis, dementia) in asymptomatic, young, menopausal women [148]. However, this view is not shared by other experts [44].

For women who use nonhormonal pharmacotherapy as their initial therapy for hot flashes, we typically taper and stop the drug after approximately two years to see if hot flashes have resolved. If hot flashes have not resolved and they are moderate to severe, the same therapy can be resumed, and another taper can be tried one to two years later. Women should be instructed that tapering SSRIs and SNRIs is important to avoid drug withdrawal symptoms.

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## SOCIETY GUIDELINE LINKS

Links to society and government-sponsored guidelines from selected countries and regions around the world are provided separately. (See ["Society guideline links: Menopause"](#).)

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## INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5<sup>th</sup> to 6<sup>th</sup> grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10<sup>th</sup> to 12<sup>th</sup> grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topics (see ["Patient education: Menopause \(The Basics\)"](#))
- Beyond the Basics topics (see ["Patient education: Menopause \(Beyond the Basics\)"](#) and ["Patient education: Menopausal hormone therapy \(Beyond the Basics\)"](#) and ["Patient education: Non-estrogen treatments for menopausal symptoms \(Beyond the Basics\)"](#) and ["Patient education: Vaginal dryness \(Beyond the Basics\)"](#))

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## SUMMARY AND RECOMMENDATIONS

- **Epidemiology** – Vasomotor symptoms (VMS) or "hot flashes" are the most common complaint during the menopausal transition, occurring in up to 80 percent of women during the late transition ( [figure 1](#)). However, only approximately 20 to 30 percent of women seek medical attention for treatment. (See ["Prevalence and natural history"](#) above.)
- **Typical symptoms** – Hot flashes typically begin as the sudden sensation of heat centered on the upper chest and face that rapidly becomes generalized. The sensation of heat lasts from two to four minutes and is often associated with profuse perspiration and occasionally palpitations (see ["Typical features"](#) above). VMS at night can cause significant sleep disturbances in many women. (See ["Effect on sleep"](#) above.)

Four different trajectories of menopausal symptoms occur with nearly equal frequency: early onset 18.4 percent, onset near final menstrual period with later decline 29.0 percent, early onset with persistently high frequency 25.6 percent, and persistently low frequency 27.0 percent ( [figure 2](#)).

- **Management**

- **Mild symptoms** – Individuals with mild hot flashes often do not seek or require pharmacologic intervention. However, some women choose nonhormonal options. (See ['Women with mild symptoms'](#) above.)
- **Moderate to severe symptoms**
  - **Hormone therapy** – For postmenopausal women with moderate to severe VMS (and no history of breast cancer or cardiovascular disease nor moderate to high risk of these diseases), we use low-dose estrogen plus progestin therapy (or estrogen alone if no uterus). Additional information on menopausal hormonal therapy (MHT), including its risks and benefits, potential candidates for therapy, as well as information on dose, regimens, and duration of therapy are discussed in detail separately. (See ["Treatment of menopausal symptoms with hormone therapy"](#), section on ['Our preferred regimens'](#) and ['Menopausal hormone therapy'](#) above.)
  - **Nonhormonal options** – For women with moderate to severe hot flashes occurring both in the daytime and at night, in whom estrogen is contraindicated, not well tolerated, or for women who have stopped estrogen and are experiencing recurrent symptoms but wish to avoid resuming estrogen, we suggest nonhormonal therapies such as selective serotonin reuptake inhibitors (SSRIs), [gabapentin](#), or [fezolinetant](#). (**Grade 2C**). (See ['Nonhormonal pharmacotherapy'](#) above.)
  - **Fezolinetant: hepatotoxicity concerns** – [Fezolinetant](#), a neurokinin 3 receptor antagonist, is another effective nonhormonal agent. However, cases of hepatotoxicity have been observed postmarketing, and FDA-revised labeling requires measurement of transaminases and bilirubin at baseline, monthly for the first three months, and at six and nine months of treatment. (See ['Neurokinin 3 receptor antagonists'](#) above.)

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## ACKNOWLEDGMENTS

The UpToDate editorial staff acknowledges Richard Santen, MD, and Charles Loprinzi, MD, who contributed to earlier versions of this topic review.

## REFERENCES

1. Randolph JF Jr, Sowers M, Bondarenko I, et al. The relationship of longitudinal change in reproductive hormones and vasomotor symptoms during the menopausal transition. *J Clin Endocrinol Metab* 2005; 90:6106.
2. National Institutes of Health. National Institutes of Health State-of-the-Science Conference statement: management of menopause-related symptoms. *Ann Intern Med* 2005; 142:1003.
3. McKinlay SM, Brambilla DJ, Posner JG. The normal menopause transition. *Maturitas* 1992; 14:103.
4. Thurston RC, Joffe H. Vasomotor symptoms and menopause: findings from the Study of Women's Health across the Nation. *Obstet Gynecol Clin North Am* 2011; 38:489.
5. McKinlay SM. The normal menopause transition: an overview. *Maturitas* 1996; 23:137.
6. Woods NF, Mitchell ES. Symptoms during the perimenopause: prevalence, severity, trajectory, and significance in women's lives. *Am J Med* 2005; 118 Suppl 12B:14.
7. Dennerstein L, Dudley EC, Hopper JL, et al. A prospective population-based study of menopausal symptoms. *Obstet Gynecol* 2000; 96:351.
8. Gold EB, Colvin A, Avis N, et al. Longitudinal analysis of the association between vasomotor symptoms and race/ethnicity across the menopausal transition: study of women's health across the nation. *Am J Public Health* 2006; 96:1226.
9. Sievert LL, Morrison L, Brown DE, Reza AM. Vasomotor symptoms among Japanese-American and European-American women living in Hilo, Hawaii. *Menopause* 2007; 14:261.
10. Crawford SL. The roles of biologic and nonbiologic factors in cultural differences in vasomotor symptoms measured by surveys. *Menopause* 2007; 14:725.
11. Soules MR, Sherman S, Parrott E, et al. Executive summary: Stages of Reproductive Aging Workshop (STRAW). *Fertil Steril* 2001; 76:874.
12. Kronenberg F. Hot flashes: epidemiology and physiology. *Ann N Y Acad Sci* 1990; 592:52.
13. Tepper PG, Brooks MM, Randolph JF Jr, et al. Characterizing the trajectories of vasomotor symptoms across the menopausal transition. *Menopause* 2016; 23:1067.
14. Grodin JM, Siiteri PK, MacDonald PC. Source of estrogen production in postmenopausal women. *J Clin Endocrinol Metab* 1973; 36:207.
15. MacDonald PC, Edman CD, Hemsell DL, et al. Effect of obesity on conversion of plasma androstenedione to estrone in postmenopausal women with and without endometrial cancer. *Am J Obstet Gynecol* 1978; 130:448.
16. Gold EB, Sternfeld B, Kelsey JL, et al. Relation of demographic and lifestyle factors to symptoms in a multi-racial/ethnic population of women 40-55 years of age. *Am J*

Epidemiol 2000; 152:463.

17. Whiteman MK, Staropoli CA, Langenberg PW, et al. Smoking, body mass, and hot flashes in midlife women. *Obstet Gynecol* 2003; 101:264.
18. Huang AJ, Subak LL, Wing R, et al. An intensive behavioral weight loss intervention and hot flushes in women. *Arch Intern Med* 2010; 170:1161.
19. Huang AJ, Grady D, Jacoby VL, et al. Persistent hot flushes in older postmenopausal women. *Arch Intern Med* 2008; 168:840.
20. Rödström K, Bengtsson C, Lissner L, et al. A longitudinal study of the treatment of hot flushes: the population study of women in Gothenburg during a quarter of a century. *Menopause* 2002; 9:156.
21. Freeman EW, Sammel MD, Sanders RJ. Risk of long-term hot flashes after natural menopause: evidence from the Penn Ovarian Aging Study cohort. *Menopause* 2014; 21:924.
22. Avis NE, Crawford SL, Greendale G, et al. Duration of menopausal vasomotor symptoms over the menopause transition. *JAMA Intern Med* 2015; 175:531.
23. Casper RF, Yen SS. Neuroendocrinology of menopausal flushes: an hypothesis of flush mechanism. *Clin Endocrinol (Oxf)* 1985; 22:293.
24. Freedman RR. Hot flashes: behavioral treatments, mechanisms, and relation to sleep. *Am J Med* 2005; 118 Suppl 12B:124.
25. Freedman RR. Physiology of hot flashes. *Am J Hum Biol* 2001; 13:453.
26. Casper RF, Yen SS, Wilkes MM. Menopausal flushes: a neuroendocrine link with pulsatile lutenizing hormone secretion. *Science* 1979; 205:823.
27. Tataryn IV, Meldrum DR, Lu KH, et al. LH, FSH and skin temperaure during the menopausal hot flash. *J Clin Endocrinol Metab* 1979; 49:152.
28. Rance NE, Dacks PA, Mittelman-Smith MA, et al. Modulation of body temperature and LH secretion by hypothalamic KNDy (kisspeptin, neurokinin B and dynorphin) neurons: a novel hypothesis on the mechanism of hot flushes. *Front Neuroendocrinol* 2013; 34:211.
29. Jayasena CN, Comninou AN, Stefanopoulou E, et al. Neurokinin B administration induces hot flushes in women. *Sci Rep* 2015; 5:8466.
30. Crandall CJ, Manson JE, Hohensee C, et al. Association of genetic variation in the tachykinin receptor 3 locus with hot flashes and night sweats in the Women's Health Initiative Study. *Menopause* 2017; 24:252.
31. Thurston RC, Sutton-Tyrrell K, Everson-Rose SA, et al. Hot flashes and subclinical cardiovascular disease: findings from the Study of Women's Health Across the Nation Heart Study. *Circulation* 2008; 118:1234.

32. Crandall CJ, Tseng CH, Crawford SL, et al. Association of menopausal vasomotor symptoms with increased bone turnover during the menopausal transition. *J Bone Miner Res* 2011; 26:840.
33. Barnabei VM, Cochrane BB, Aragaki AK, et al. Menopausal symptoms and treatment-related effects of estrogen and progestin in the Women's Health Initiative. *Obstet Gynecol* 2005; 105:1063.
34. Santoro N, Braunstein GD, Butts CL, et al. Compounded Bioidentical Hormones in Endocrinology Practice: An Endocrine Society Scientific Statement. *J Clin Endocrinol Metab* 2016; 101:1318.
35. Freedman RR, Roehrs TA. Effects of REM sleep and ambient temperature on hot flash-induced sleep disturbance. *Menopause* 2006; 13:576.
36. Erlik Y, Tataryn IV, Meldrum DR, et al. Association of waking episodes with menopausal hot flushes. *JAMA* 1981; 245:1741.
37. Ohayon MM. Severe hot flashes are associated with chronic insomnia. *Arch Intern Med* 2006; 166:1262.
38. de Zambotti M, Colrain IM, Javitz HS, Baker FC. Magnitude of the impact of hot flashes on sleep in perimenopausal women. *Fertil Steril* 2014; 102:1708.
39. Kravitz HM, Janssen I, Santoro N, et al. Relationship of day-to-day reproductive hormone levels to sleep in midlife women. *Arch Intern Med* 2005; 165:2370.
40. Williams RE, Levine KB, Kalilani L, et al. Menopause-specific questionnaire assessment in US population-based study shows negative impact on health-related quality of life. *Maturitas* 2009; 62:153.
41. New Collective Author. The 2023 nonhormone therapy position statement of The North American Menopause Society. *Menopause* 2023; 30:573.
42. Ziaei S, Kazemnejad A, Zareai M. The effect of vitamin E on hot flashes in menopausal women. *Gynecol Obstet Invest* 2007; 64:204.
43. Barton DL, Loprinzi CL, Quella SK, et al. Prospective evaluation of vitamin E for hot flashes in breast cancer survivors. *J Clin Oncol* 1998; 16:495.
44. Stuenkel CA, Davis SR, Gompel A, et al. Treatment of Symptoms of the Menopause: An Endocrine Society Clinical Practice Guideline. *J Clin Endocrinol Metab* 2015; 100:3975.
45. "The 2022 Hormone Therapy Position Statement of The North American Menopause Society" Advisory Panel. The 2022 hormone therapy position statement of The North American Menopause Society. *Menopause* 2022; 29:767.
46. North American Menopause Society. Treatment of menopause-associated vasomotor symptoms: position statement of The North American Menopause Society. *Menopause* 2004; 11:11.

47. Landgren MB, Bennink HJ, Helmond FA, Engelen S. Dose-response analysis of effects of tibolone on climacteric symptoms. *BJOG* 2002; 109:1109.
48. Pinkerton JV, Santen R. Alternatives to the use of estrogen in postmenopausal women. *Endocr Rev* 1999; 20:308.
49. Modelska K, Cummings S. Tibolone for postmenopausal women: systematic review of randomized trials. *J Clin Endocrinol Metab* 2002; 87:16.
50. Evans K, Colloca L, Pecina M, Katz N. What can be done to control the placebo response in clinical trials? A narrative review. *Contemp Clin Trials* 2021; 107:106503.
51. Loprinzi CL, Sloan J, Stearns V, et al. Newer antidepressants and gabapentin for hot flashes: an individual patient pooled analysis. *J Clin Oncol* 2009; 27:2831.
52. Nelson HD, Vesco KK, Haney E, et al. Nonhormonal therapies for menopausal hot flashes: systematic review and meta-analysis. *JAMA* 2006; 295:2057.
53. Sloan JA, Loprinzi CL, Novotny PJ, et al. Methodologic lessons learned from hot flash studies. *J Clin Oncol* 2001; 19:4280.
54. Rada G, Capurro D, Pantoja T, et al. Non-hormonal interventions for hot flushes in women with a history of breast cancer. *Cochrane Database Syst Rev* 2010; :CD004923.
55. Sideras K, Loprinzi CL. Nonhormonal management of hot flashes for women on risk reduction therapy. *J Natl Compr Canc Netw* 2010; 8:1171.
56. Loprinzi CL, Kugler JW, Sloan JA, et al. Venlafaxine in management of hot flashes in survivors of breast cancer: a randomised controlled trial. *Lancet* 2000; 356:2059.
57. Stearns V, Beebe KL, Iyengar M, Dube E. Paroxetine controlled release in the treatment of menopausal hot flashes: a randomized controlled trial. *JAMA* 2003; 289:2827.
58. Loprinzi CL, Sloan JA, Perez EA, et al. Phase III evaluation of fluoxetine for treatment of hot flashes. *J Clin Oncol* 2002; 20:1578.
59. Evans ML, Pritts E, Vittinghoff E, et al. Management of postmenopausal hot flushes with venlafaxine hydrochloride: a randomized, controlled trial. *Obstet Gynecol* 2005; 105:161.
60. Stearns V, Slack R, Greep N, et al. Paroxetine is an effective treatment for hot flashes: results from a prospective randomized clinical trial. *J Clin Oncol* 2005; 23:6919.
61. Loprinzi CL, Barton DL, Sloan JA, et al. Mayo Clinic and North Central Cancer Treatment Group hot flash studies: a 20-year experience. *Menopause* 2008; 15:655.
62. Barton DL, LaVasseur BI, Sloan JA, et al. Phase III, placebo-controlled trial of three doses of citalopram for the treatment of hot flashes: NCCTG trial N05C9. *J Clin Oncol* 2010; 28:3278.
63. Freeman EW, Guthrie KA, Caan B, et al. Efficacy of escitalopram for hot flashes in healthy menopausal women: a randomized controlled trial. *JAMA* 2011; 305:267.

64. Guttuso T Jr, Kurlan R, McDermott MP, Kieburtz K. Gabapentin's effects on hot flashes in postmenopausal women: a randomized controlled trial. *Obstet Gynecol* 2003; 101:337.
65. Pandya KJ, Morrow GR, Roscoe JA, et al. Gabapentin for hot flashes in 420 women with breast cancer: a randomised double-blind placebo-controlled trial. *Lancet* 2005; 366:818.
66. Bardia A, Novotny P, Sloan J, et al. Efficacy of nonestrogenic hot flash therapies among women stratified by breast cancer history and tamoxifen use: a pooled analysis. *Menopause* 2009; 16:477.
67. Simon JA, Portman DJ, Kaunitz AM, et al. Low-dose paroxetine 7.5 mg for menopausal vasomotor symptoms: two randomized controlled trials. *Menopause* 2013; 20:1027.
68. Suvanto-Luukkonen E, Koivunen R, Sundström H, et al. Citalopram and fluoxetine in the treatment of postmenopausal symptoms: a prospective, randomized, 9-month, placebo-controlled, double-blind study. *Menopause* 2005; 12:18.
69. Kalay AE, Demir B, Haberal A, et al. Efficacy of citalopram on climacteric symptoms. *Menopause* 2007; 14:223.
70. LaCroix AZ, Freeman EW, Larson J, et al. Effects of escitalopram on menopause-specific quality of life and pain in healthy menopausal women with hot flashes: a randomized controlled trial. *Maturitas* 2012; 73:361.
71. Cronin-Fenton DP, Damkier P, Lash TL. Metabolism and transport of tamoxifen in relation to its effectiveness: new perspectives on an ongoing controversy. *Future Oncol* 2014; 10:107.
72. Kerwin JP, Gordon PR, Senf JH. The variable response of women with menopausal hot flashes when treated with sertraline. *Menopause* 2007; 14:841.
73. Grady D, Cohen B, Tice J, et al. Ineffectiveness of sertraline for treatment of menopausal hot flushes: a randomized controlled trial. *Obstet Gynecol* 2007; 109:823.
74. Joffe H, Guthrie KA, LaCroix AZ, et al. Low-dose estradiol and the serotonin-norepinephrine reuptake inhibitor venlafaxine for vasomotor symptoms: a randomized clinical trial. *JAMA Intern Med* 2014; 174:1058.
75. Bordeleau L, Pritchard KI, Loprinzi CL, et al. Multicenter, randomized, cross-over clinical trial of venlafaxine versus gabapentin for the management of hot flashes in breast cancer survivors. *J Clin Oncol* 2010; 28:5147.
76. Speroff L, Gass M, Constantine G, et al. Efficacy and tolerability of desvenlafaxine succinate treatment for menopausal vasomotor symptoms: a randomized controlled trial. *Obstet Gynecol* 2008; 111:77.
77. Loprinzi L, Barton DL, Sloan JA, et al. Pilot evaluation of gabapentin for treating hot flashes. *Mayo Clin Proc* 2002; 77:1159.
78. Fitzpatrick LA, Santen RJ. Hot flashes: the old and the new, what is really true? *Mayo Clin Proc* 2002; 77:1155.

79. Reddy SY, Warner H, Guttuso T Jr, et al. Gabapentin, estrogen, and placebo for treating hot flashes: a randomized controlled trial. *Obstet Gynecol* 2006; 108:41.
80. Loprinzi CL, Kugler JW, Barton DL, et al. Phase III trial of gabapentin alone or in conjunction with an antidepressant in the management of hot flashes in women who have inadequate control with an antidepressant alone: NCCTG N03C5. *J Clin Oncol* 2007; 25:308.
81. Loprinzi CL, Qin R, Balcueva EP, et al. Phase III, randomized, double-blind, placebo-controlled evaluation of pregabalin for alleviating hot flashes, N07C1. *J Clin Oncol* 2010; 28:641.
82. Leon-Ferre RA, Novotny PJ, Faubion SS, et al. Abstract GS6-02: A randomized, double-blind, placebo-controlled trial of oxybutynin (Oxy) for hot flashes (HF): ACCRU study SC-1603. *Cancer Res* 2019; 79:GS6.
83. Simon JA, Gaines T, LaGuardia KD, Extended-Release Oxybutynin Therapy for VMS Study Group. Extended-release oxybutynin therapy for vasomotor symptoms in women: a randomized clinical trial. *Menopause* 2016; 23:1214.
84. Skorupskaite K, George JT, Veldhuis JD, et al. Neurokinin 3 Receptor Antagonism Reveals Roles for Neurokinin B in the Regulation of Gonadotropin Secretion and Hot Flashes in Postmenopausal Women. *Neuroendocrinology* 2018; 106:148.
85. Johnson KA, Martin N, Nappi RE, et al. Efficacy and Safety of Fezolinetant in Moderate to Severe Vasomotor Symptoms Associated With Menopause: A Phase 3 RCT. *J Clin Endocrinol Metab* 2023; 108:1981.
86. Lederman S, Ottery FD, Cano A, et al. Fezolinetant for treatment of moderate-to-severe vasomotor symptoms associated with menopause (SKYLIGHT 1): a phase 3 randomised controlled study. *Lancet* 2023; 401:1091.
87. Veozah: Highlights of prescribing information. US Food and Drug Administration. [https://www.accessdata.fda.gov/drugsatfda\\_docs/label/2024/216578s004lbl.pdf](https://www.accessdata.fda.gov/drugsatfda_docs/label/2024/216578s004lbl.pdf) (Accessed on December 17, 2024).
88. Simon JA, Anderson RA, Ballantyne E, et al. Efficacy and safety of elinzanetant, a selective neurokinin-1,3 receptor antagonist for vasomotor symptoms: a dose-finding clinical trial (SWITCH-1). *Menopause* 2023; 30:239.
89. Pinkerton JV, Simon JA, Joffe H, et al. Elinzanetant for the Treatment of Vasomotor Symptoms Associated With Menopause: OASIS 1 and 2 Randomized Clinical Trials. *JAMA* 2024; 332:1343.
90. Cardoso F, Parke S, Brennan DJ, et al. Elinzanetant for Vasomotor Symptoms from Endocrine Therapy for Breast Cancer. *N Engl J Med* 2025; 393:753.
91. Jin Y, Hayes DF, Li L, et al. Estrogen receptor genotypes influence hot flash prevalence and composite score before and after tamoxifen therapy. *J Clin Oncol* 2008; 26:5849.

92. Jones SE, Cantrell J, Vukelja S, et al. Comparison of menopausal symptoms during the first year of adjuvant therapy with either exemestane or tamoxifen in early breast cancer: report of a Tamoxifen Exemestane Adjuvant Multicenter trial substudy. *J Clin Oncol* 2007; 25:4765.
93. Santen RJ, Stuenkel CA, Davis SR, et al. Managing Menopausal Symptoms and Associated Clinical Issues in Breast Cancer Survivors. *J Clin Endocrinol Metab* 2017; 102:3647.
94. Noehr-Jensen L, Zwisler ST, Larsen F, et al. Escitalopram is a weak inhibitor of the CYP2D6-catalyzed O-demethylation of (+)-tramadol but does not reduce the hypoalgesic effect in experimental pain. *Clin Pharmacol Ther* 2009; 86:626.
95. Newton KM, Buist DS, Keenan NL, et al. Use of alternative therapies for menopause symptoms: results of a population-based survey. *Obstet Gynecol* 2002; 100:18.
96. Keenan NL, Mark S, Fugh-Berman A, et al. Severity of menopausal symptoms and use of both conventional and complementary/alternative therapies. *Menopause* 2003; 10:507.
97. McCurry SM, Guthrie KA, Morin CM, et al. Telephone-Based Cognitive Behavioral Therapy for Insomnia in Perimenopausal and Postmenopausal Women With Vasomotor Symptoms: A MsFLASH Randomized Clinical Trial. *JAMA Intern Med* 2016; 176:913.
98. Mann E, Smith M, Hellier J, Hunter MS. A randomised controlled trial of a cognitive behavioural intervention for women who have menopausal symptoms following breast cancer treatment (MENOS 1): trial protocol. *BMC Cancer* 2011; 11:44.
99. Ayers B, Smith M, Hellier J, et al. Effectiveness of group and self-help cognitive behavior therapy in reducing problematic menopausal hot flushes and night sweats (MENOS 2): a randomized controlled trial. *Menopause* 2012; 19:749.
100. Ensrud KE, Guthrie KA, Hohensee C, et al. Effects of estradiol and venlafaxine on insomnia symptoms and sleep quality in women with hot flashes. *Sleep* 2015; 38:97.
101. Santoro N, Epperson CN, Mathews SB. Menopausal Symptoms and Their Management. *Endocrinol Metab Clin North Am* 2015; 44:497.
102. Elkins G, Marcus J, Stearns V, et al. Randomized trial of a hypnosis intervention for treatment of hot flashes among breast cancer survivors. *J Clin Oncol* 2008; 26:5022.
103. Elkins GR, Fisher WI, Johnson AK, et al. Clinical hypnosis in the treatment of postmenopausal hot flashes: a randomized controlled trial. *Menopause* 2013; 20:291.
104. Nedrow A, Miller J, Walker M, et al. Complementary and alternative therapies for the management of menopause-related symptoms: a systematic evidence review. *Arch Intern Med* 2006; 166:1453.
105. Saensak S, Vutyavanich T, Somboonporn W, Srisurapanont M. Relaxation for perimenopausal and postmenopausal symptoms. *Cochrane Database Syst Rev* 2014; :CD008582.

106. Carmody JF, Crawford S, Salmoirago-Blotcher E, et al. Mindfulness training for coping with hot flashes: results of a randomized trial. *Menopause* 2011; 18:611.
107. Walega DR, Rubin LH, Banuvar S, et al. Effects of stellate ganglion block on vasomotor symptoms: findings from a randomized controlled clinical trial in postmenopausal women. *Menopause* 2014; 21:807.
108. Othman AH, Zaky AH. Management of hot flushes in breast cancer survivors: comparison between stellate ganglion block and pregabalin. *Pain Med* 2014; 15:410.
109. Kronenberg F, Fugh-Berman A. Complementary and alternative medicine for menopausal symptoms: a review of randomized, controlled trials. *Ann Intern Med* 2002; 137:805.
110. Van Patten CL, Olivotto IA, Chambers GK, et al. Effect of soy phytoestrogens on hot flashes in postmenopausal women with breast cancer: a randomized, controlled clinical trial. *J Clin Oncol* 2002; 20:1449.
111. Upmalis DH, Lobo R, Bradley L, et al. Vasomotor symptom relief by soy isoflavone extract tablets in postmenopausal women: a multicenter, double-blind, randomized, placebo-controlled study. *Menopause* 2000; 7:236.
112. Quella SK, Loprinzi CL, Barton DL, et al. Evaluation of soy phytoestrogens for the treatment of hot flashes in breast cancer survivors: A North Central Cancer Treatment Group Trial. *J Clin Oncol* 2000; 18:1068.
113. Nikander E, Kilkkinen A, Metsä-Heikkilä M, et al. A randomized placebo-controlled crossover trial with phytoestrogens in treatment of menopause in breast cancer patients. *Obstet Gynecol* 2003; 101:1213.
114. MacGregor CA, Canney PA, Patterson G, et al. A randomised double-blind controlled trial of oral soy supplements versus placebo for treatment of menopausal symptoms in patients with early breast cancer. *Eur J Cancer* 2005; 41:708.
115. Levis S, Strickman-Stein N, Ganjei-Azar P, et al. Soy isoflavones in the prevention of menopausal bone loss and menopausal symptoms: a randomized, double-blind trial. *Arch Intern Med* 2011; 171:1363.
116. Franco OH, Chowdhury R, Troup J, et al. Use of Plant-Based Therapies and Menopausal Symptoms: A Systematic Review and Meta-analysis. *JAMA* 2016; 315:2554.
117. Butt DA, Deng LY, Lewis JE, Lock M. Minimal decrease in hot flashes desired by postmenopausal women in family practice. *Menopause* 2007; 14:203.
118. Tice JA, Ettinger B, Ensrud K, et al. Phytoestrogen supplements for the treatment of hot flashes: the Isoflavone Clover Extract (ICE) Study: a randomized controlled trial. *JAMA* 2003; 290:207.
119. Lethaby A, Marjoribanks J, Kronenberg F, et al. Phytoestrogens for menopausal vasomotor symptoms. *Cochrane Database Syst Rev* 2013; :CD001395.

120. Frei-Kleiner S, Schaffner W, Rahlfs VW, et al. *Cimicifuga racemosa* dried ethanolic extract in menopausal disorders: a double-blind placebo-controlled clinical trial. *Maturitas* 2005; 51:397.
121. Osmer R, Friede M, Liske E, et al. Efficacy and safety of isopropanolic black cohosh extract for climacteric symptoms. *Obstet Gynecol* 2005; 105:1074.
122. Verhoeven MO, van der Mooren MJ, van de Weijer PH, et al. Effect of a combination of isoflavones and *Actaea racemosa* Linnaeus on climacteric symptoms in healthy symptomatic perimenopausal women: a 12-week randomized, placebo-controlled, double-blind study. *Menopause* 2005; 12:412.
123. Pockaj BA, Gallagher JG, Loprinzi CL, et al. Phase III double-blind, randomized, placebo-controlled crossover trial of black cohosh in the management of hot flashes: NCCTG Trial N01CC1. *J Clin Oncol* 2006; 24:2836.
124. Newton KM, Reed SD, LaCroix AZ, et al. Treatment of vasomotor symptoms of menopause with black cohosh, multibotanicals, soy, hormone therapy, or placebo: a randomized trial. *Ann Intern Med* 2006; 145:869.
125. Leach MJ, Moore V. Black cohosh (*Cimicifuga* spp.) for menopausal symptoms. *Cochrane Database Syst Rev* 2012; :CD007244.
126. Roberts H. Safety of herbal medicinal products in women with breast cancer. *Maturitas* 2010; 66:363.
127. Fritz H, Seely D, McGowan J, et al. Black cohosh and breast cancer: a systematic review. *Integr Cancer Ther* 2014; 13:12.
128. L'Espérance S, Frenette S, Dionne A, et al. Pharmacological and non-hormonal treatment of hot flashes in breast cancer survivors: CEPO review and recommendations. *Support Care Cancer* 2013; 21:1461.
129. Joy D, Joy J, Duane P. Black cohosh: a cause of abnormal postmenopausal liver function tests. *Climacteric* 2008; 11:84.
130. Naser B, Schnitker J, Minkin MJ, et al. Suspected black cohosh hepatotoxicity: no evidence by meta-analysis of randomized controlled clinical trials for isopropanolic black cohosh extract. *Menopause* 2011; 18:366.
131. Zhu X, Liew Y, Liu ZL. Chinese herbal medicine for menopausal symptoms. *Cochrane Database Syst Rev* 2016; 3:CD009023.
132. Freedman RR, Woodward S. Behavioral treatment of menopausal hot flushes: evaluation by ambulatory monitoring. *Am J Obstet Gynecol* 1992; 167:436.
133. Sood R, Sood A, Wolf SL, et al. Paced breathing compared with usual breathing for hot flashes. *Menopause* 2013; 20:179.
134. Deng G, Vickers A, Yeung S, et al. Randomized, controlled trial of acupuncture for the treatment of hot flashes in breast cancer patients. *J Clin Oncol* 2007; 25:5584.

135. Wyon Y, Wijma K, Nedstrand E, Hammar M. A comparison of acupuncture and oral estradiol treatment of vasomotor symptoms in postmenopausal women. *Climacteric* 2004; 7:153.
136. Cho SH, Whang WW. Acupuncture for vasomotor menopausal symptoms: a systematic review. *Menopause* 2009; 16:1065.
137. Lee MS, Kim KH, Choi SM, Ernst E. Acupuncture for treating hot flashes in breast cancer patients: a systematic review. *Breast Cancer Res Treat* 2009; 115:497.
138. Hervik J, Mjåland O. Acupuncture for the treatment of hot flashes in breast cancer patients, a randomized, controlled trial. *Breast Cancer Res Treat* 2009; 116:311.
139. Dodin S, Blanchet C, Marc I, et al. Acupuncture for menopausal hot flushes. *Cochrane Database Syst Rev* 2013; :CD007410.
140. Ee C, Xue C, Chondros P, et al. Acupuncture for Menopausal Hot Flashes: A Randomized Trial. *Ann Intern Med* 2016; 164:146.
141. Mao JJ, Bowman MA, Xie SX, et al. Electroacupuncture Versus Gabapentin for Hot Flashes Among Breast Cancer Survivors: A Randomized Placebo-Controlled Trial. *J Clin Oncol* 2015; 33:3615.
142. Chenoy R, Hussain S, Tayob Y, et al. Effect of oral gamolenic acid from evening primrose oil on menopausal flushing. *BMJ* 1994; 308:501.
143. Dodin S, Lemay A, Jacques H, et al. The effects of flaxseed dietary supplement on lipid profile, bone mineral density, and symptoms in menopausal women: a randomized, double-blind, wheat germ placebo-controlled clinical trial. *J Clin Endocrinol Metab* 2005; 90:1390.
144. Pruthi S, Qin R, Terstreip SA, et al. A phase III, randomized, placebo-controlled, double-blind trial of flaxseed for the treatment of hot flashes: North Central Cancer Treatment Group N08C7. *Menopause* 2012; 19:48.
145. Aiello EJ, Yasui Y, Tworoger SS, et al. Effect of a yearlong, moderate-intensity exercise intervention on the occurrence and severity of menopause symptoms in postmenopausal women. *Menopause* 2004; 11:382.
146. Daley A, Stokes-Lampard H, Thomas A, MacArthur C. Exercise for vasomotor menopausal symptoms. *Cochrane Database Syst Rev* 2014; :CD006108.
147. Lindh-Astrand L, Nedstrand E, Wyon Y, Hammar M. Vasomotor symptoms and quality of life in previously sedentary postmenopausal women randomised to physical activity or estrogen therapy. *Maturitas* 2004; 48:97.
148. Lobo RA. Hormone-replacement therapy: current thinking. *Nat Rev Endocrinol* 2017; 13:220.

## GRAPHICS

Table 1: Causes of flushing

| Type        | Causes                                                                                                                                                                                                                                                                                                                                                                     |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Physiologic | <ul style="list-style-type: none"><li>▪ Menopause</li><li>▪ Hot drinks</li><li>▪ Emotional distress</li><li>▪ Anaphylaxis</li></ul>                                                                                                                                                                                                                                        |
| Drugs       | <ul style="list-style-type: none"><li>▪ Alcohol</li><li>▪ Alcohol plus chlorpromazine or disulfiram</li><li>▪ Diltiazem</li><li>▪ Amyl nitrate</li><li>▪ Nicotinic acid (niacin)</li><li>▪ Levodopa</li><li>▪ Bromocriptine</li></ul>                                                                                                                                      |
| Diseases    | <ul style="list-style-type: none"><li>▪ Carcinoid syndrome</li><li>▪ Systemic mastocytosis</li><li>▪ Basophilic chronic granulocytic leukemia</li><li>▪ VIPoma</li><li>▪ Pheochromocytoma</li><li>▪ Medullary carcinoma of the thyroid</li><li>▪ Renal cell carcinoma</li><li>▪ Diencephalic seizures</li><li>▪ Postural orthostatic tachycardia syndrome (POTS)</li></ul> |

VIPoma: vasoactive intestinal peptide secreting tumor.

Figure 1: The Stages of Reproductive Aging Workshop +10 staging system for reproductive aging in females

|                                      |                     |         |            |                                  |                                                                               |                                             |                                |                                    |                                           |      |
|--------------------------------------|---------------------|---------|------------|----------------------------------|-------------------------------------------------------------------------------|---------------------------------------------|--------------------------------|------------------------------------|-------------------------------------------|------|
| Menarche                             |                     |         |            |                                  | FMP (0)                                                                       |                                             |                                |                                    |                                           |      |
| Stage                                | -5                  | -4      | -3b        | -3a                              | -2                                                                            | -1                                          | +1a                            | +1b                                | +1c                                       | +2   |
| Terminology                          | REPRODUCTIVE        |         |            |                                  | MENOPAUSAL TRANSITION                                                         |                                             | POSTMENOPAUSE                  |                                    |                                           |      |
|                                      | Early               | Peak    | Late       |                                  | Early                                                                         | Late                                        | Early                          |                                    |                                           | Late |
|                                      |                     |         |            |                                  | Perimenopause                                                                 |                                             |                                |                                    |                                           |      |
| Duration                             | Variable            |         |            |                                  | Variable                                                                      | 1-3 years                                   | 2 years (1+1)                  | 3-6 years                          | Remaining lifespan                        |      |
| PRINCIPAL CRITERIA                   |                     |         |            |                                  |                                                                               |                                             |                                |                                    |                                           |      |
| Menstrual cycle                      | Variable to regular | Regular | Regular    | Subtle changes in flow/ strength | Variable length: Persistent ≥7-day difference in length of consecutive cycles | Interval of amenorrhea of ≥60 days          |                                |                                    |                                           |      |
| SUPPORTIVE CRITERIA                  |                     |         |            |                                  |                                                                               |                                             |                                |                                    |                                           |      |
| Endocrine<br>FSH<br>AMH<br>Inhibin B |                     |         | Low<br>Low | Variable*<br>Low<br>Low          | ↑ Variable*<br>Low<br>Low                                                     | ↑ >25 international units/L ¶<br>Low<br>Low | ↑ Variable<br>Low<br>Low       | Stabilizes<br>Very low<br>Very low |                                           |      |
| Antral follicle count                |                     |         | Low        | Low                              | Low                                                                           | Low                                         | Very low                       | Very low                           |                                           |      |
| DESCRIPTIVE CHARACTERISTICS          |                     |         |            |                                  |                                                                               |                                             |                                |                                    |                                           |      |
| Symptoms                             |                     |         |            |                                  |                                                                               | Vasomotor symptoms likely                   | Vasomotor symptoms most likely |                                    | Increasing symptoms of urogenital atrophy |      |

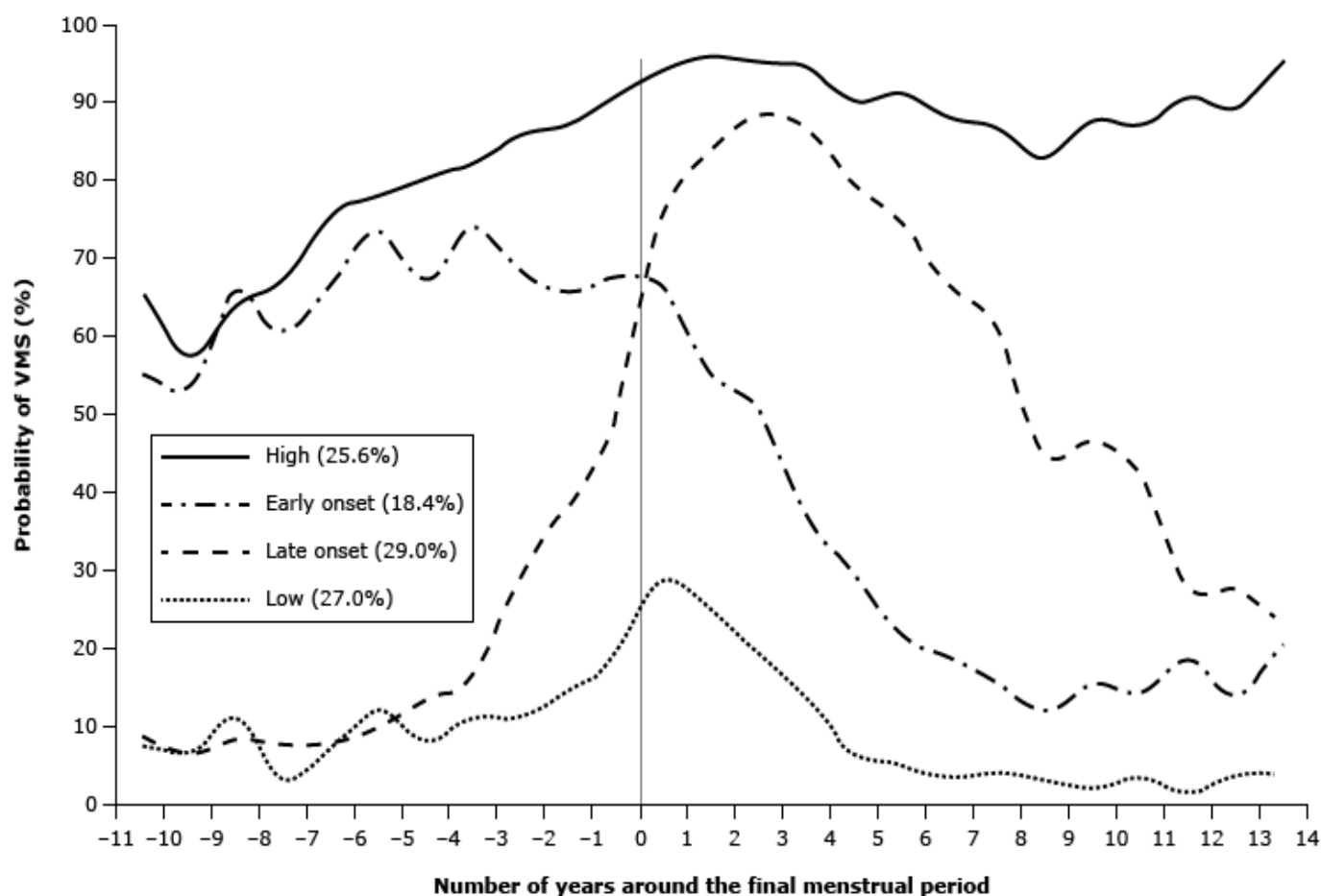
FMP: final menstrual period; FSH: follicle-stimulating hormone; AMH: anti-müllerian hormone; Arrow: elevated.

\* Blood draw on cycle days 2 to 5.

¶ Approximate expected level based on assays using current international pituitary standard.

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Figure 2: Trajectories of vasomotor symptoms over the menopause transition



Probability of VMS represents the average observed probability of VMS at each time point within each trajectory subgroup. No factors were included in the model.

VMS: vasomotor symptoms.

From: Tepper PG, Brooks MM, Randolph JF, et al. Characterizing the trajectories of vasomotor symptoms across the menopausal transition. *Menopause* 2016 23:1067. DOI: [10.1097/GME.0000000000000676](https://doi.org/10.1097/GME.0000000000000676). Copyright © 2016 The North American Menopause Society. Reproduced with permission from Lippincott Williams & Wilkins. Unauthorized reproduction of this material is prohibited.

Table 2: Medications that may cause sweating or flushing

|                                            |                               |
|--------------------------------------------|-------------------------------|
| <b>Antidepressants</b>                     | <b>Hypoglycemic agents</b>    |
| Bupropion                                  | Insulin                       |
| SSRIs                                      | Sulfonylureas                 |
| Tricyclic antidepressants                  | Thiazolidinediones            |
| SNRIs                                      | <b>Sympathomimetic agents</b> |
| <b>Antimigraine drugs</b>                  | Beta-agonists                 |
| "Triptans" serotonin 5-HT (1b/1d) agonists | Phenylephrine                 |
| <b>Antipyretics</b>                        | <b>Miscellaneous</b>          |
| Acetaminophen                              | Alcohol                       |
| Aspirin                                    | Beta blockers                 |
| NSAIDs                                     | Bromocriptine                 |
| <b>Cholinergic agonists</b>                | Calcium channel blockers      |
| Bethanechol                                | Clozapine                     |
| Pilocarpine                                | Cyclosporine                  |
| Pyridostigmine                             | Donepezil                     |
| <b>Hormonal agents</b>                     | Fluvoxamine                   |
| <b>GnRH agonists</b>                       | Hydralazine                   |
| Gonadorelin                                | Imatinib                      |
| Goserelin                                  | Infliximab                    |
| Histrelin                                  | Morphine                      |
| Leuprolide                                 | Niacin                        |
| Nafarelin                                  | Nitroglycerin                 |
| <b>Aromatase inhibitors</b>                | Omeprazole                    |
| Anastrozole                                | Opioids                       |
| Exemestane                                 | Protease inhibitors           |
| Letrozole                                  | Rituximab                     |
| <b>Receptor blockers</b>                   | Ropinirole                    |
| Flutamide                                  | Sildenafil                    |
| <b>SERMs</b>                               | Theophylline                  |
| Raloxifene                                 | Tramadol                      |
| Tamoxifen                                  |                               |

SSRIs: selective serotonin reuptake inhibitors; SNRIs: serotonin-norepinephrine reuptake inhibitors; NSAIDs: nonsteroidal antiinflammatory drugs; GnRH: gonadotropin-releasing hormone; SERMs: selective estrogen receptor modulators.

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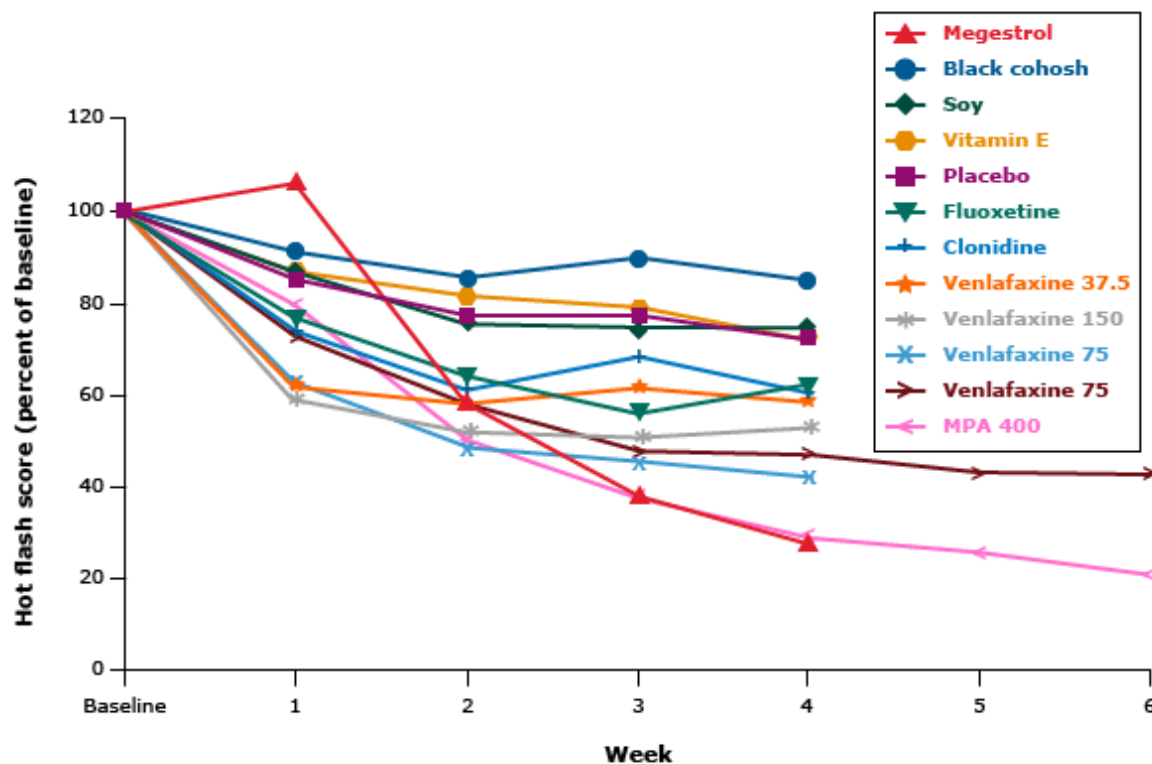
Graphic 56351 Version 8.0

Table 3: Differential diagnosis of night sweats or flushing

| <b>Malignancy</b>                  | <b>Endocrine disorders</b>   |
|------------------------------------|------------------------------|
| Lymphoma                           | Carcinoid syndrome           |
| Non-Hodgkin lymphoma               | Diabetes insipidus           |
| Hodgkin lymphoma                   | Hyperthyroidism              |
| Solid tumors                       | Hypoglycemia                 |
| Germ cell tumors                   | Pheochromocytoma             |
| Medullary carcinoma of the thyroid | Postorchiectomy              |
| Prostate cancer                    | <b>Menopause</b>             |
| Renal cell carcinoma               | <b>Neurologic disorders</b>  |
| Miscellaneous cancers              | Autonomic dysreflexia        |
| <b>Infections</b>                  | Autonomic neuropathy         |
| Mycobacterial                      | Post-traumatic syringomyelia |
| Tuberculosis                       | Stroke                       |
| Atypical mycobacteria              | <b>Miscellaneous</b>         |
| Bacterial                          | Chronic fatigue syndrome     |
| Abscess                            | Common blushing              |
| Brucellosis                        | Food additives               |
| Endocarditis                       | Gastroesophageal reflux      |
| Osteomyelitis                      | Mastocytosis                 |
| Fungal infections                  | Panic disorder               |
| Viral                              | Rosacea                      |
| HIV infection                      | Sleep apnea                  |
| Chronic hepatitis C                | Temporal arteritis           |
| <b>Medications</b>                 | Idiopathic hyperhidrosis     |
| <b>Substance withdrawal</b>        |                              |
| Alcohol                            |                              |
| Cocaine                            |                              |
| Opioids                            |                              |

HIV: human immunodeficiency virus.

Figure 3: Therapies hot flashes



Hot flash score changes from baseline for a series of 8 randomized, placebo-controlled, double-blind clinical trials in women. Hot flash score changes from baseline for the same trials illustrated, plus a 6-week trial in which women were randomized to venlafaxine 75 mg versus a single dose of intramuscular medroxyprogesterone acetate 400 mg.

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## Contributor Disclosures

**Robert F Casper, MD** Stocks: TRIO Fertility [Infertility]. Consultant/Advisory Boards: Abbott International [Infertility]; AbbVie [Endometriosis]; EMD-Serono [In vitro fertilization]; Ferring Pharmaceuticals [In vitro fertilization]. All of the relevant financial relationships listed have been mitigated. **Robert L Barbieri, MD** No relevant financial relationship(s) with ineligible companies to disclose. **Jean E Mulder, MD** No relevant financial relationship(s) with ineligible companies to disclose.

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